

CLINICAL INVESTIGATION

Eye

OCULAR RESPONSE OF CHOROIDAL MELANOMA WITH MONOSOMY 3 VERSUS DISOMY 3 AFTER IODINE-125 BRACHYTHERAPY

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Purpose: To report the ocular response of choroidal melanoma with monosomy 3 vs. disomy 3 after ¹²⁵I brachytherapy.

Methods and Materials: We evaluated patients with ciliochoroidal melanoma managed with fine needle aspiration biopsy immediately before plaque application for ¹²⁵I brachytherapy between January 1, 2005 and December 31, 2008. Patients with (1) cytopathologic diagnosis of melanoma, (2) melanoma chromosome 3 status identified by fluorescence *in situ* hybridization, and (3) 6 or more months of follow-up after brachytherapy were sorted by monosomy 3 vs. disomy 3 and compared by Kruskal-Wallis test.

Results: Among 40 ciliochoroidal melanomas (40 patients), 15 had monosomy 3 and 25 had disomy 3. Monosomy 3 melanomas had a median greatest basal diameter of 12.00 mm and a median tumor thickness of 6.69 mm before brachytherapy; at a median of 1.75 years after brachytherapy, median thickness was 3.10 mm. Median percentage decrease in tumor thickness was 48.3%. Disomy 3 melanomas had a median greatest basal diameter of 10.00 mm and median tumor thickness of 3.19 mm before brachytherapy; at a median of 2.00 years after brachytherapy, median tumor thickness was 2.37 mm. The median percentage decrease in tumor thickness was 22.7%. Monosomy 3 melanomas were statistically greater in size than disomy 3 melanomas ($p < 0.001$) and showed a greater decrease in tumor thickness after brachytherapy ($p = 0.006$).

Conclusion: In this study, ciliochoroidal melanomas with monosomy 3 were significantly greater in size than disomy 3 melanoma and showed a significantly greater decrease in thickness at a median of 1.75 years after brachytherapy. The greater decrease in monosomy 3 melanoma thickness after brachytherapy is consistent with other malignancies in which more aggressive pathology has been shown to be associated with a greater initial response to radiotherapy. © 2011 Elsevier Inc.

Ciliochoroidal melanoma, Choroidal melanoma, Uveal melanoma, Monosomy 3, ¹²⁵I brachytherapy.

INTRODUCTION

Melanoma arising from the ciliary body and the choroid is the most common primary intraocular malignancy of adults. The Collaborative Ocular Melanoma Study randomized clinical trial of ¹²⁵I brachytherapy vs. enucleation for medium-sized melanoma showed that, for patients who met eligibility criteria, there was no statistically significant difference in all-cause mortality between ¹²⁵I brachytherapy and enucleation 5, 10, and 12 years after treatment (1). Therefore, ¹²⁵I brachytherapy alone is the standard of care for globe-sparing local treatment of ciliochoroidal melanoma.

Testable with cytogenetic studies, loss of one copy of chromosome 3, or monosomy 3, in tumor tissue is detected in approximately half of ciliochoroidal melanomas and is the risk factor most strongly associated with metastatic death

(2, 3). Studies of ciliochoroidal melanoma have shown that patients with an early rapid tumor response to radiation have a greater likelihood to develop metastatic disease (4–6). It remains unclear whether the finding of monosomy 3 would necessarily predict a differential degree of local tumor response as compared with tumors without monosomy 3.

In this study, we sought to evaluate the response of ciliochoroidal melanoma with known chromosome 3 status after treatment with ¹²⁵I brachytherapy.

METHODS AND MATERIALS

A retrospective review of all patients with a clinical diagnosis of ciliochoroidal melanoma who underwent ¹²⁵I brachytherapy and evaluation of chromosome 3 status by fluorescence *in situ*

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